EXPERIMENT – 5

Recover Deleted Files using Forensics Tools (FTK Imager)

Aim

To recover a deleted file from a storage device by creating a forensically sound bit-by-bit image using FTK Imager, and to verify the integrity of the image using hash values.

Procedure

- Step 1 (Create the file): Create a sample file 'malicious.txt' in Notepad, type some content into it and save it to the drive.
- Step 2 (Delete the file): Delete the file so that it is no longer visible to the user. The data of the deleted file still remains in the unallocated space of the drive until it is overwritten.
- Step 3 (Understand imaging): A deleted file is not copied by an ordinary file copy or backup. Therefore, a forensic investigator first makes a forensically sound bit-by-bit image of the drive.
- Step 4 (Start the imager): Install and open FTK Imager, and select File → Create Disk Image.
- Step 5 (Select the source): In the Select Source dialog, choose the evidence type – Physical Drive (a logical drive or an existing image file can also be selected) – and click 'Next'.
- Step 6 (Choose the image type): In the Select Destination Image Type dialog, choose the image format – Raw (dd), SMART, E01 or AFF. E01 is preferred because it supports compression and integrity verification.
- Step 7 (Evidence details): Enter the Evidence Item Information – case number, evidence number, unique description, the examiner name and any notes.
- Step 8 (Destination): Specify the Image Destination Folder, the image file name, the fragment size (MB) and the compression level.
- Step 9 (Create the image): Click 'Start'. FTK Imager begins creating the bit-by-bit image of the drive and shows the progress.

- Step 10 (Verify the image): After the image is created, use File → Verify Drive/Image to compute and compare the MD5 and SHA-1 hash values and confirm that the image is an exact copy of the source.
- Step 11 (Recover the deleted file): Analyse the resulting image so that the deleted file can be recovered (data carving), and compare the recovered content with the original to confirm the recovery.

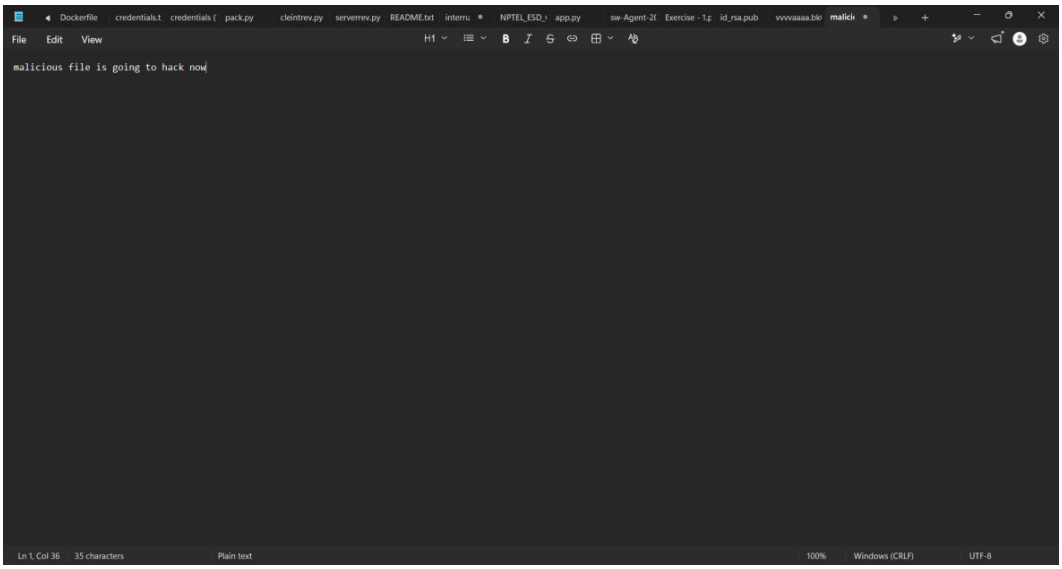


Fig. 5.1 Creating the target file 'malicious.txt' with the sample content
Timestamp: 12-10-2026, 15:05:12

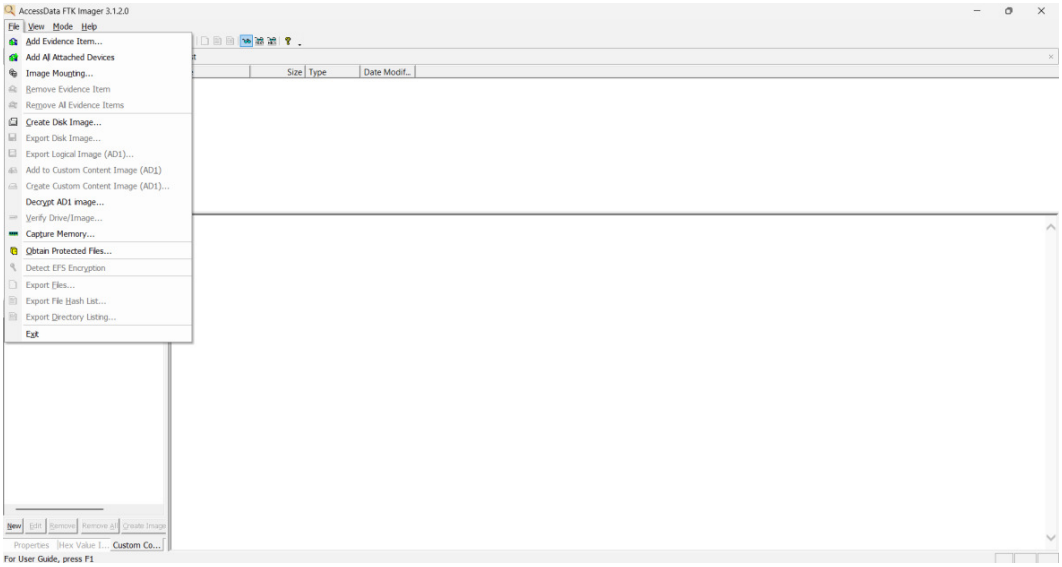


Fig. 5.2 FTK Imager – selecting 'Create Disk Image' from the File menu
Timestamp: 12-10-2026, 15:06:35

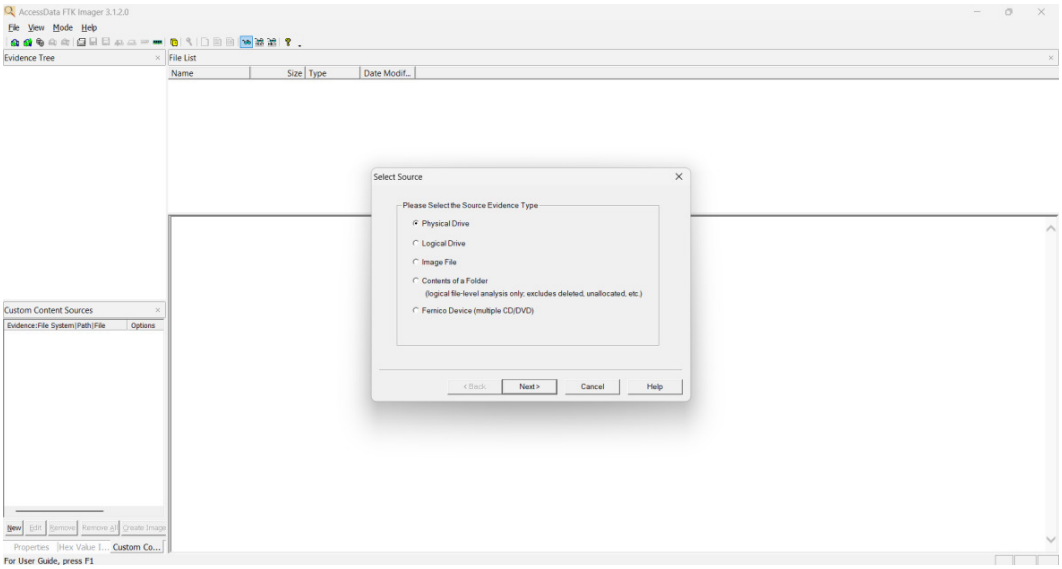


Fig. 5.3 FTK Imager – selecting the source evidence type
Timestamp: 12-10-2026, 15:12:06

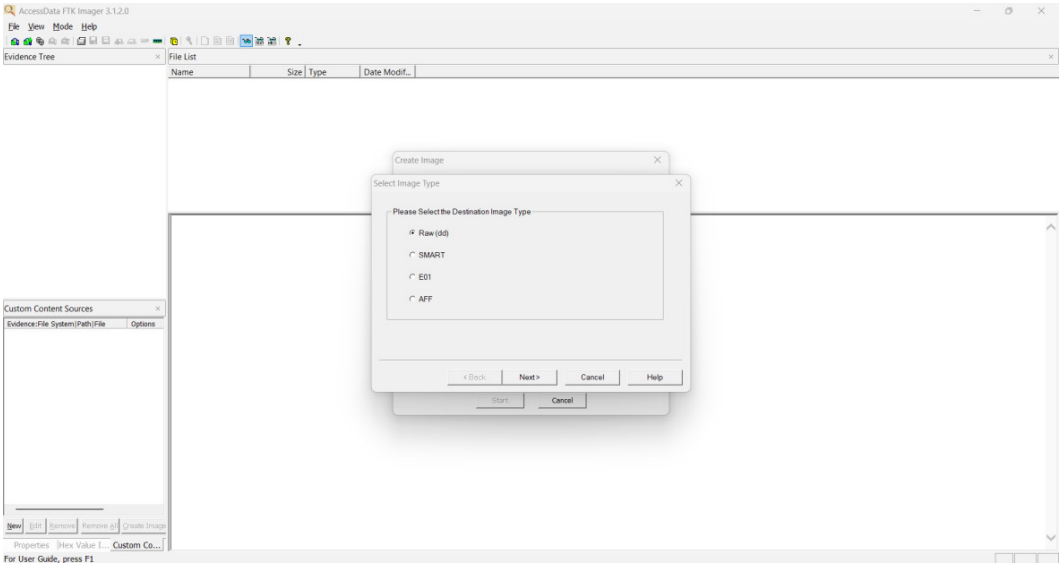


Fig. 5.4 FTK Imager – choosing the destination image type (Raw dd)
Timestamp: 12-10-2026, 15:12:20

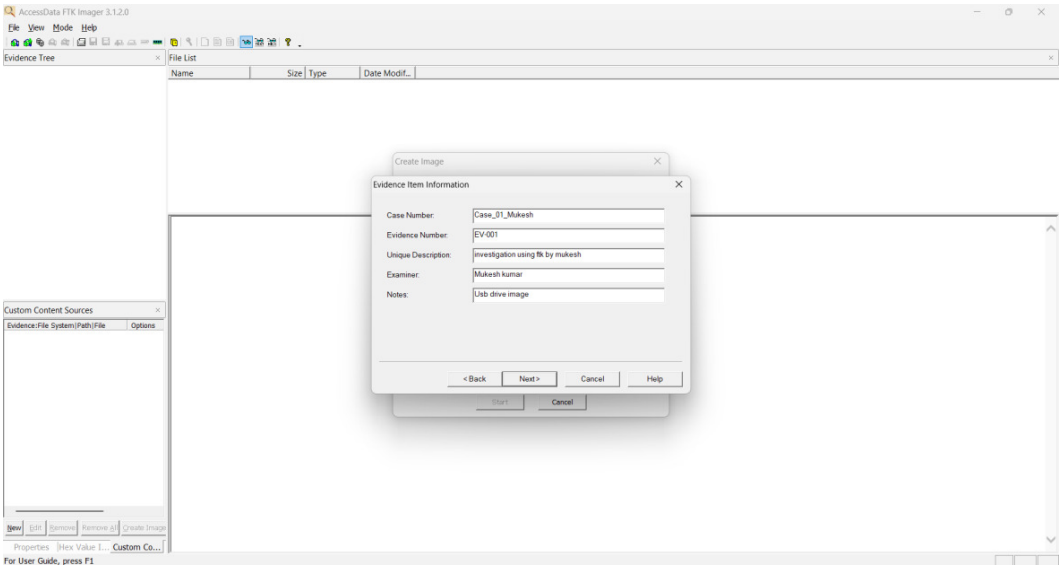


Fig. 5.5 FTK Imager – entering the evidence item information
Timestamp: 12-10-2026, 15:14:01

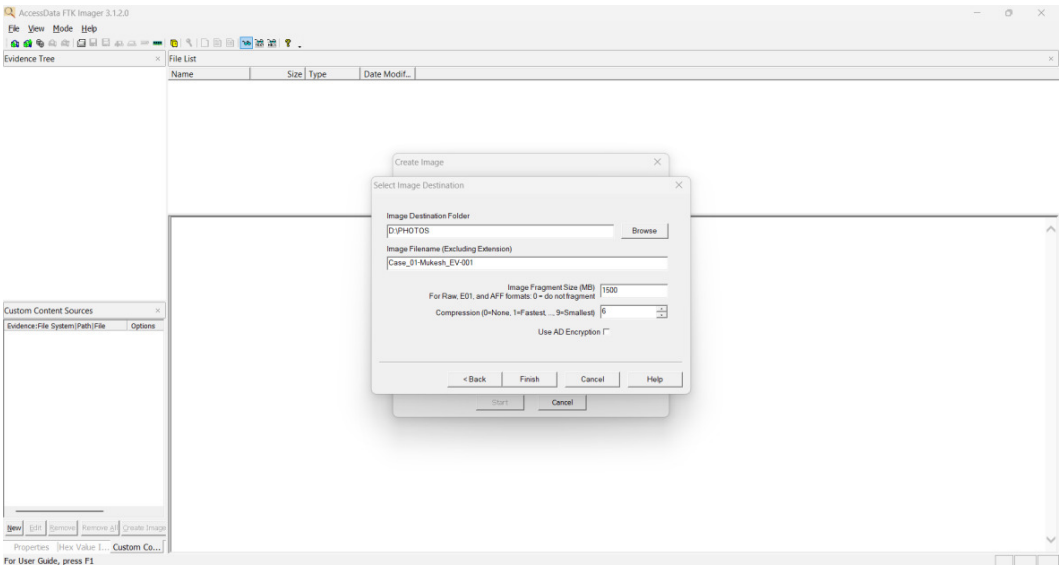


Fig. 5.6 FTK Imager – specifying the image destination folder and file name
Timestamp: 12-10-2026, 15:15:06

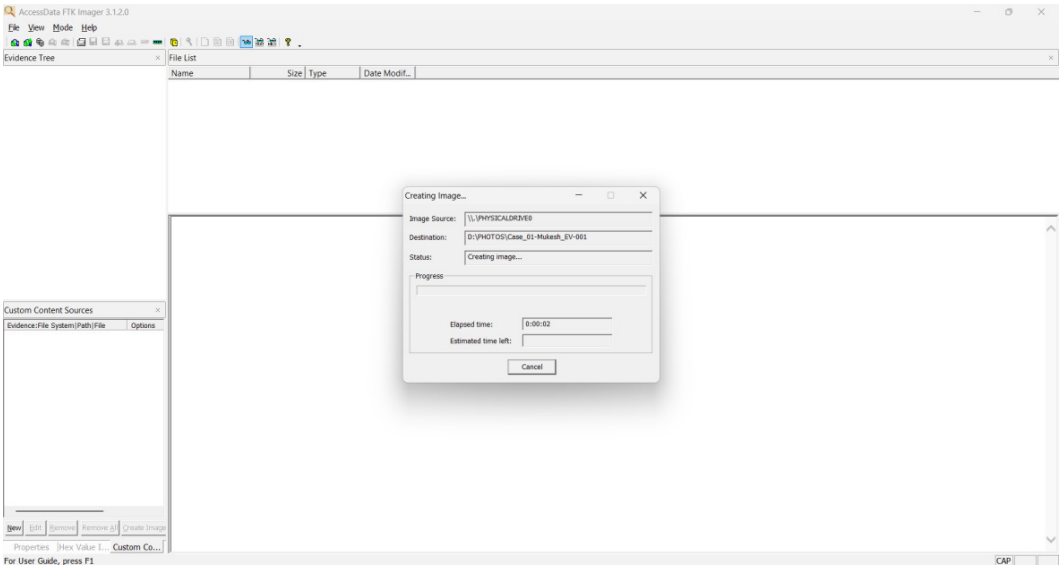


Fig. 5.7 FTK Imager – creating the bit-by-bit image of the drive
Timestamp: 12-10-2026, 15:15:25

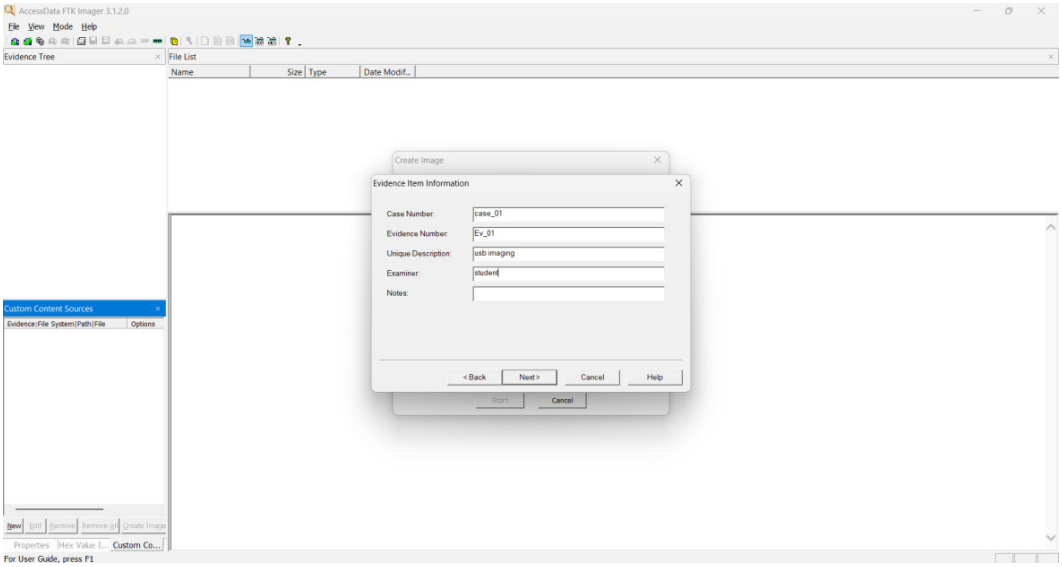


Fig. 5.8 FTK Imager – entering the evidence item information for the device image
Timestamp: 14-10-2026, 19:46:44

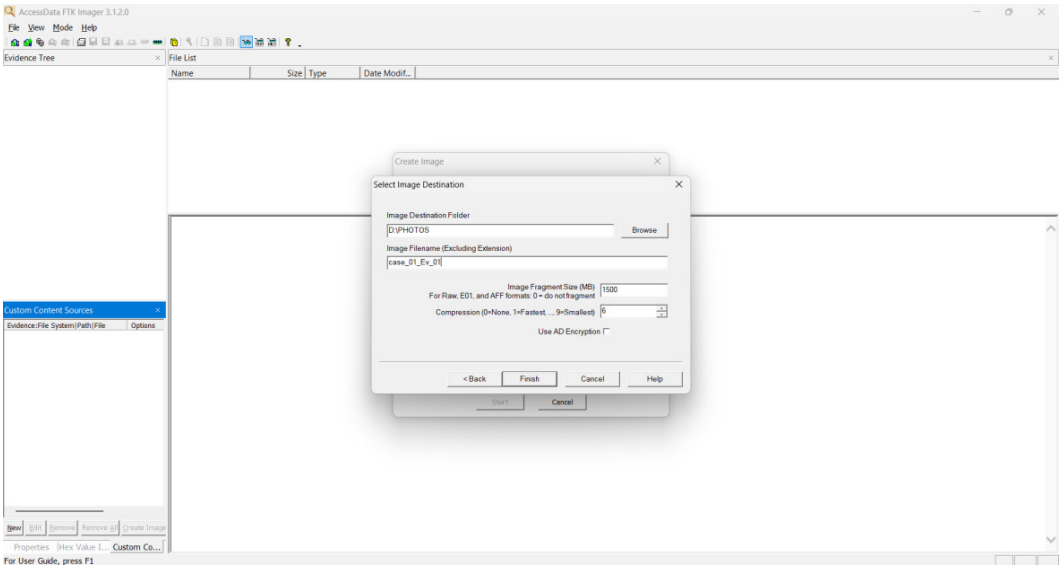


Fig. 5.9 FTK Imager – setting the image destination folder
Timestamp: 14-10-2026, 19:47:16

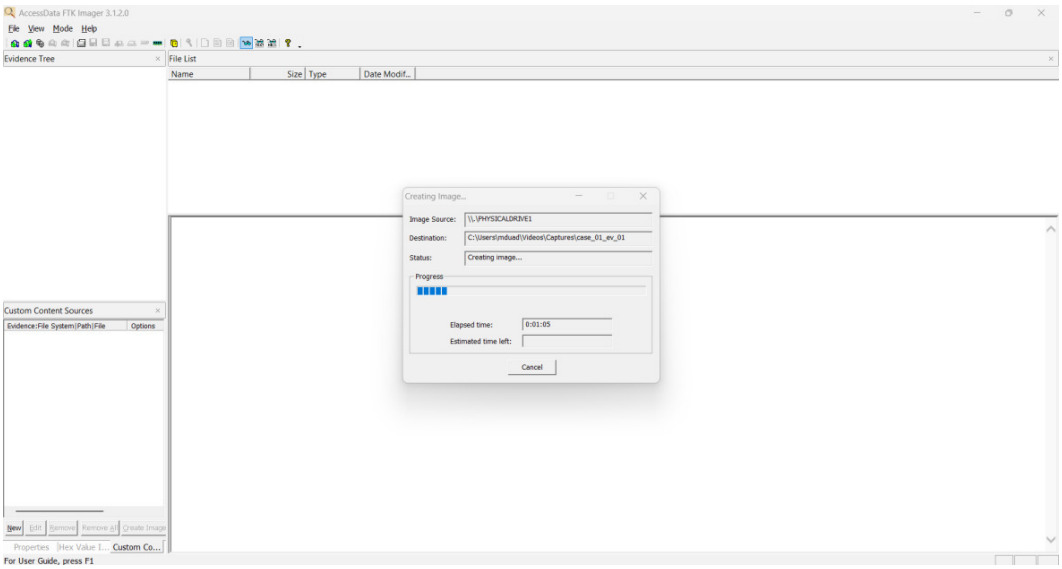


Fig. 5.10 FTK Imager – imaging the device (creating the image)
Timestamp: 14-10-2026, 19:50:49

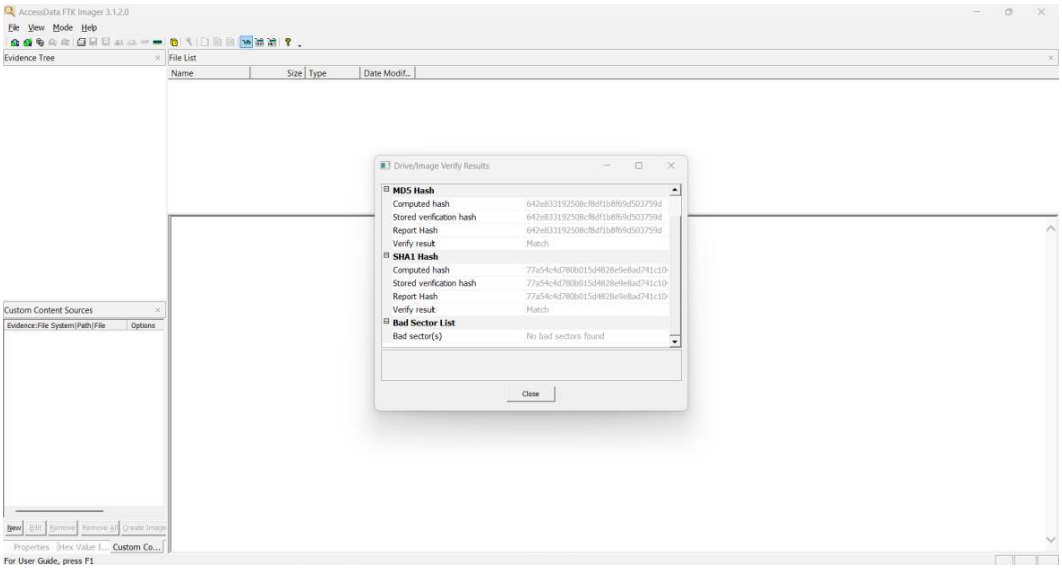


Fig. 5.11 FTK Imager – verifying the image (MD5 and SHA-1 hash values)
Timestamp: 14-10-2026, 20:02:58

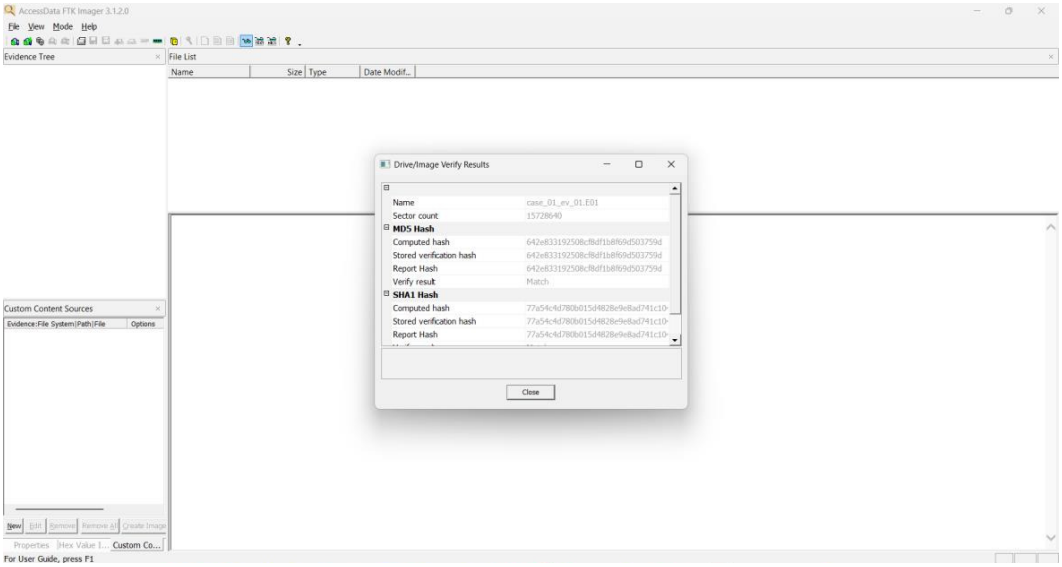


Fig. 5.12 FTK Imager – drive/image verification results
Timestamp: 14-10-2026, 20:03:12

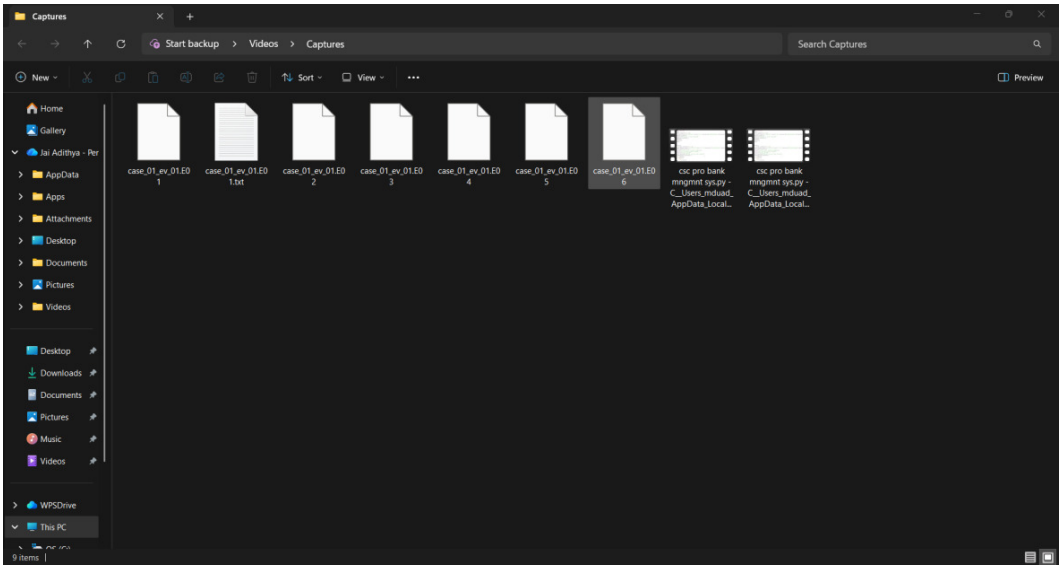


Fig. 5.13 The generated E01 image segments of the device
Timestamp: 14-10-2026, 20:56:41

Result

Thus, the deleted file was recovered using forensic tools. A bit-by-bit image of the device was created with FTK Imager, the integrity of the image was verified using the MD5 and SHA-1 hash values, and the deleted content was recovered and verified successfully.